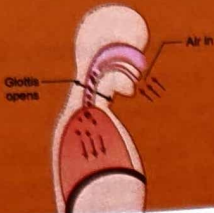


CHAPTER

14



Kasa

Learning Objectives

- ⇒ To learn the concept of Kasa
- ⇒ To learn the Nidana and Samprapti of Kasa mentioned in Ayurveda literatures
- ⇒ To frame out the probable Samprapti Ghataka based on the progression of the disease
- ⇒ To know the types of Kasa
- ⇒ To know the Purvarupa of Kasa with current understanding
- ⇒ To differentiate the types of Kasa based on Dosha dominance
- ⇒ To learn about the Sadhyasadyata of Kasa with current understanding
- ⇒ To learn the differential diagnosis of Kasa based on the Lakshana of Kapha
- ⇒ To learn the differential diagnosis of Kshayaja Kasa and Rajayakshma
- ⇒ To learn the differential diagnosis of Kshataja Kasa and Kshata Kshina

Introduction

Kasa¹ is the spectrum of diseases where we can observe the involvement *Pranavaha Srotas*. This is characterised by explosive expiration where in the irritating contents present in *Pranavaha Srotas* is being eliminated. This is manifested in an individual due to abnormal function of *Pranavata* and *Udanavata*. Depending upon the dominance of *Dosha* during the progression, we can observe the dominance of *Vata* (dry cough) or *Kapha* (productive cough). *Sanga* and *Vimargagamana* are observed in this disease. This is categorised under *Sharirika Vikara* and *Amashayasamutthta Vikara* of *Doshabala Pravritta Vikara* of *Adhyatmika Vikara*.

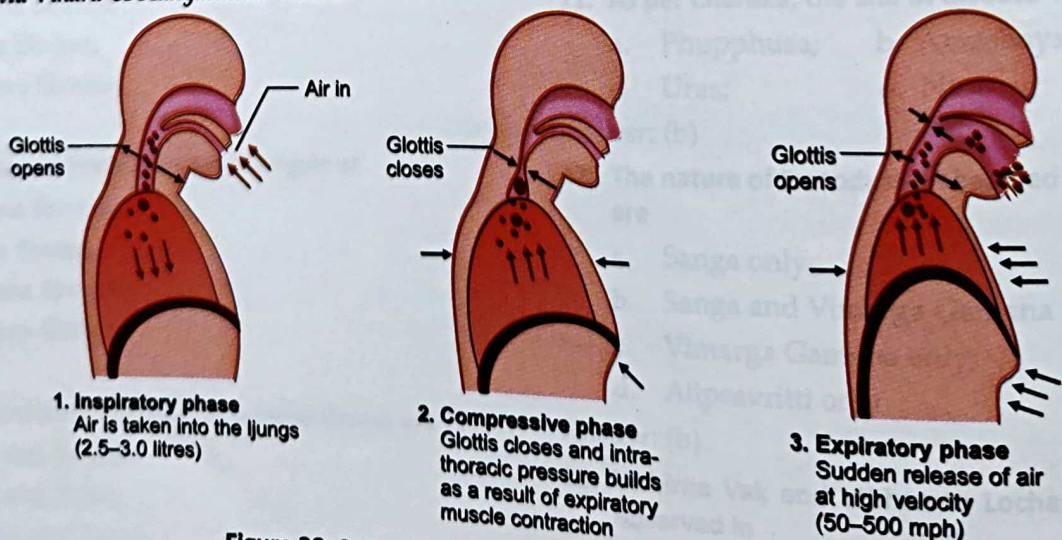


Figure 28: Samprapti of Kasa (mechanism of cough)

Derivation of Kasa

प्राणो ह्युदानानुगतः प्रदुष्टः संभिन्न कांस्यस्वनतुल्यघोषः।
निरेति वक्रात्सहसा सदोपः कासः स विद्वद्भिर्मुदाहृतस्तु॥ Su.Ut. – 52/4

कसनात्कास उच्यते

Ca.Ci. – 18/8

This disease is characterised by the production of specific type of sound i.e. sound produced by tapping the broken bronze vessel. This is facilitated by the combined action of *Prana* and *Udana Vata*.

Nidana of Kasa

कसनात्कास उच्यते (धूमोपघाताद्रजसस्तथैव व्यायाम-
रूक्षान्निषेवणाच्च।

वेगावरोधात् क्ष्वथोस्तथैव) विभागगत्वादिपि भोजनस्य

Su.Ut. – 52/3, Ma.Ni. – 11/1

विदाहिगुरुविष्टम्भिरूक्षाभिष्यन्दिभोजनैः।

शीतपानासनस्थानरजोधूमानिलानलैः॥3॥

व्यायाम कर्मभाराध्ववेगाघातापतर्पणैः।

आमदोषाभिघातस्त्रीक्षयरोगप्रपीडनैः॥4॥

विषमाशनाध्यनशनैस्तथा संशमनैरपि।

ह्रिक्का श्वासश्च कासश्च नृणां समुपजायते॥5॥ Su.Ut. – 50/3-5

⊗ Exposure to smoke

⊗ Exposure to dust²

- ⊗ Excess performance of exercise³
- ⊗ Excess intake of dry food substances⁴ such as *Chanaka* (chick pea) etc
- ⊗ Displacement of ingested food moving into the *Pranavaha Srotas*⁵
- ⊗ Suppression of natural urges, sneezing⁶ in particular
- ⊗ Consumption of *Vidahi*⁷ (responsible for the burning sensation in chest and thirst) food substances in excess such as *Maricha* (black pepper), *Sarshapa* (mustard seeds) or alcoholic beverages
- ⊗ Excess consumption of *Guru* (heavy to digest) substances⁸
- ⊗ Excess consumption of *Vishtambhi*⁹ (substances leading to poor movement in the alimentary canal) *Ahara* such as *jackfruit* or *Karira* – *Capparis aphylla*, a thorny plant grown in deserts eaten by camels
- ⊗ Excess consumption of *Abhishyandi*¹⁰ (substances responsible for *Srotorodha*) *Ahara* such as consumption of fish, blackgram and curds
- ⊗ Exposure of the individual to cold atmosphere such as cold drinks intake, staying in air conditioned room, cold air exposure etc¹¹
- ⊗ Excess performance of physically debilitating exercise
- ⊗ Lifting heavyweight more than one's capacity

- 2 Each nostril is having certain amount of hairs at the entrance to prevent the movement of smoke and dust particles. The second phase of defence is presence of turbinates in the nasal cavity. The third phase of defence is cilia present in the respiratory mucosa and the fourth phase of defence is mucus secreted by the mucous membranes of respiratory system. These are different levels of defence which prevent the entry of dust and smoke particles of various size which are usually less than 6 micrometers of size. Particles which are of smaller than 0.5 micrometers in diameter are usually suspended in alveolar air and are expelled by expiration. Cigarette smoke particles are usually of 0.3 micrometer and 1/3rd of these particles are precipitated in the alveoli and rest of the particles are suspended in the air and expelled out via expired air.
These particles are considered as irritants to the respiratory airways and thus they are responsible for the manifestation of cough.
- 3 When an individual performs strenuous exercise, that will increase the air hunger in the individual. This air hunger makes the individual to take the deep breath through open mouth. During this time, the person may have the entry of dust or other particles enter into the respiratory passages, leading to the irritation and inflammation of upper and lower respiratory passages, with the presenting complaint of cough.
- 4 As per fundamentals of Ayurveda, any substance consumed responsible for the vitiation of Vata Dosha, may have an impact on *Pranavaha Srotas* also leading to the manifestation of Kasa. As per current understanding, the person who has consumed dry food substances, that will lead to the stimulation of parasympathetic nervous system initially, in due course of the time, there may be a possibility of parasympathetic nervous system is stimulated, in the alimentary canal, there is increased secretion of juices. In nervous system. When the parasympathetic nervous stimulation, respiratory secretions are increased which initiate the act of coughing. respiratory system, during the parasympathetic nervous stimulation, respiratory secretions are increased which initiate the act of coughing.
- 5 When ingested food displaces into the *Pranavaha Srotas* (respiratory passages), that will lead to aspiration pneumonia with presenting complaints of productive cough, fever, breathlessness.
- 6 Sneezing is a protective reflex mechanism which becomes active when an individual is exposed to dust, smoke or any respiratory allergen either through nose or mouth. If a man suppresses this reflex, then the dust, smoke or respiratory allergen likely to travel from the upper respiratory tract to lower respiratory tract, leading to the clinical presentation of cough, breathlessness, fever etc,
- 7 Consumption of these substances such as black pepper, mustard seeds or alcoholic beverages will lead to the vitiation of Pitta and Vata Dosha in the body.
- 8 The consumption of these food substances are responsible for the vitiation of Kapha Dosha by default. These substances will lead to the increase of *Snigdha*, *Shita*, *Guru*, *Shlakshna* qualities in the body and this increase in the respiratory system will be responsible for the appearance of breathlessness in the suffering individual.
- 9 The excess consumption of *Vishtambhi* substances will lead to the abnormal increase of Vata Dosha in the body.
- 10 The excess consumption of *Abhishyandi* substances leads to the excess secretion in alimentary canal, respiratory system. In the respiratory system, it causes irritation of the respiratory system leading to the manifestation of cough, breathlessness etc.
- 11 When an individual is exposed to cold atmosphere, that will lead to the dryness of the respiratory mucosa and in this situation, the dust, smoke or allergen directly attach to the respiratory mucosa leading to cough. In the trachea and bronchi, that will lead to the broncho constriction and constriction of trachea and this will lead to breathlessness and cough.

- ❖ Excess walking¹²
- ❖ Suppression of natural urges¹³
- ❖ Doing the measures which lead to depletion of body tissues
- ❖ If the person is having presence of Ama in the body
- ❖ If the person is performing excess sexual activity¹⁴
- ❖ If the person is suffering from other chronic debilitating illness¹⁵
- ❖ If the person is having the habit of consuming the food at improper time and improper quantity¹⁶
- ❖ If the person is consuming the food before the digestion of previously ingested meals¹⁷

Samprapti of Kasa

- अधःप्रतिहतो वायुरूर्ध्वं स्रोतः समाश्रितः।
 उदानभावमापन्नः कण्ठे सक्तस्तथोरसि॥6॥
 आविश्य शिरसः खानि सर्वाणि प्रतिपूरयन्।
 आभञ्जन्नाक्षिपन् देहं हनुमन्ये तथाऽक्षिणी॥7॥
 नेत्रे पृष्ठमुरः पार्श्वे निर्भुज्य स्तम्भयन्स्ततः।
 शुष्को वा सकफो वाऽपि कसनात्कास उच्यते॥8॥ Ca.Ci. – 30/6-8
- ❖ The vitiated Dosha from downwards gets further

vitiated due to above said etiological factors and tend to move upwards towards direction of Pranavaha Srotas.

- ❖ This vitiated Vata will move towards Uras (chest) and Kantha (throat) and comes forcefully through mouth producing a typical sound.
- ❖ During this process, there will be contraction of chest muscles, abdomen, neck, face, back making the body bend forwards during the act of coughing.
- ❖ There can be either dry or productive cough.

Samprapti Ghataka

- ❖ Dosha : Kapha and Vata (Prana and Udana)
- ❖ Dushya : Rasa
- ❖ Srotas : Pranavaha and Rasavaha Srotas
- ❖ Srotodushti Prakara : Sanga and Vimargagamana
- ❖ Agni : Jatharagnimandya
- ❖ Ama : Jatharagnimandyaajanya
- ❖ Udbhava Sthana : Amashaya
- ❖ Sanchara Sthana : Sarva Sharira (as Rasa Dhatu involvement is present) and Uras region (chest)

- 12 It's a question mark, how excess exercise can lead to the occurrence of breathlessness. In the textbook of physiology written by Guyton and Hall, normal oxygen consumption for a young man at rest is 250 ml/ minute. This quantity may increase to 3600 ml/ minute in untrained average male; athletically trained male increases upto 4000 ml/minute and male marathon runner, oxygen consumption may increase to 5100 ml/ minute.
 As far as pulmonary ventilation is concerned, at rest pulmonary ventilation is occurring about 9 to 11 litres/ minute and during maximal exercise, the pulmonary ventilation increases about 100 to 110 litres/ minute and maximal breathing capacity increases about 150 to 170 litres/ minute.
 When the person is performing exercise with such maximal exercise, the person will also breathe through open mouth. The other determinants of changes in pulmonary ventilation are exercise at higher altitudes and exercise under very hot conditions and pre existing respiratory disease.
 This will lead to the entry of numerous respiratory irritants and microbes into the respiratory passages leading to the obstruction to respiratory passages.
- 13 Let us take the example of urge of defaecation. When an individual is having the urge of defaecation and he suppresses the urge will lead to the vitiation of Apanavata in Pakvashaya and this will lead to the reverse of Apanavata and this mechanism is termed as Udavarta. In due course of time, upward movement of Apanavata afflicts Samanavata and Pranavata sequentially. Thus, breathlessness can be observed as a late feature in suppression of defaecation.
- 14 When an individual is susceptible to respiratory disease, and if the individual is indulged in sexual intercourse, then the person will have air hunger, this will lead to breathing through open mouth and the person may have direct entry of dust, smoke or allergen and the person develops cough.
- 15 When an individual is suffering from either cardiac disease or upper alimentary canal disease may usually develop respiratory in due course of time.
- 16 Let us take the example of consumption of food substances at large quantity at prior to the digestion of previously ingested meal, this may lead to poor or delayed digestion. This may lead to metabolic acidosis either ketoacidosis or lactic acidosis or both. The further state of this acidosis is respiratory alkalosis. This is characterised by breathing discomfort.
 - Suppose if the person consumes very less amount of food at a particular point of time, it will lead to sequential depletion of carbohydrate, fats and proteins. This may lead to state of metabolic acidosis (ketoacidosis via fat metabolism and lactic acidosis via metabolism in the muscles). Further may develop respiratory alkalosis.
 - If the person takes the food substances too early, it may act as Adhyashana or Atyashana (too much amount of food), this may act like effect produced i.e. prolonged digestion of food, which may lead to metabolic acidosis and later there will be respiratory alkalosis.
- 17 The term Adhyashana refers to ingestion of meals before the digestion of previously ingested meal. As per fundamentals of Ayurveda, there will be vitiation of Kapha Dosha along with dominance of Vata Dosha.
 When an individual is consuming excess intake of food before the digestion, that will lead to the accumulation of partially digested foods in the storage. This will lead to reverse peristaltic movement leading to aspiration pneumonia if gastric contents enter the respiratory passages, thus leading to the mechanisms of cough and breathlessness.

- Vyakta Sthaana : Vaktra
- Rogamarga : Abhyantara and Bahya

Purvarupa of Kasa

पूर्वरूपं भवेत्तेषां शूक पूर्णगलास्यता।
कण्ठे कण्डूश्च भोज्यानामवरोधश्च जायते॥5॥ Ca.Ci. – 18/5
भविष्यतस्तस्य तु कण्ठकण्डूभोज्योपरोधो गलतालुलेपः।
शब्दवैषम्यमरोचकोऽग्निसादश्च लिङ्गानि भवन्त्यमूनि॥6॥

Su.Ut. – 52/6

- Sensation of pricking of throat and oral cavity by thorns¹⁸
- Itching in the throat¹⁹
- Sense of obstruction of food in the throat²⁰
- Coating in the throat and palate²¹
- Hoarseness of voice²²
- Tastelessness²³
- Loss of appetite²⁴

Types of Kasa

5 types of Kasa: as per Charaka, Sushruta and Madhavakara

1. Vataja Kasa
2. Pittaja Kasa
3. Kaphaja Kasa
4. Kshataja Kasa
5. Kshayaja Kasa

Vataja Kasa

रूक्षशीतकषयाल्पप्रमितानशनं स्त्रियः।
वेगधारणमायासो वातकासप्रवर्तकाः॥10॥
हृत्पाश्चीरः शिरः शूलस्वरभेदकरो भृशम्।
शुष्कोरः कण्ठवक्रस्य हृष्टलोमः प्रताम्यतः॥11॥
निर्घोष दैन्यस्तनन दौर्बल्यक्षोभमोहकृत्।
शुष्ककासः कफं शुष्कं कृच्छ्रान्मुक्त्वाऽल्पतां व्रजेत् स्निग्धाप्लव
वणोष्णैश्च भुक्तपीतैः प्रशाम्यति।
ऊर्ध्ववातस्य जीर्णेने वेगवान्मारुतो भवेत्॥13॥ Ca.Ci. – 18/10-13
हृच्छङ्गमूर्धोदरपार्श्वशूली क्षामाननः क्षीणबल खरौजाः।
प्रसक्तमन्तः कफमीरणेन कासेत्तु शुष्कं स्वरभेदयुक्तः॥17॥
हृच्छङ्गमूर्धोदरपार्श्वशूली क्षामाननः क्षीणबलस्वरौजाः।
प्रसक्तवेगस्तु समीरणेन भिन्नस्वरः कासति शुष्कमेव॥15॥
Ma.Ni. – 11/5

Su.Ut. – 52/7

- Consumption of dry substances in excess
- Consumption of cold substances in excess
- Consumption of food substances having astringent taste
- Consumption of little amount of food
- Consumption of food with single Rasa dominance
- Excess indulgence in sexual intercourse
- Suppression of natural urges
- Excess performance of physically debilitating activities

Lakshana of Vataja Kasa²⁵

- Pricking type of pain in chest, sides of the chest,

G
K

- 18 Sensation of pricking may indicate the ongoing inflammatory changes in the throat and oral cavity. When there is inflammatory reaction, that will lead to the protective mechanism in order to remove the cause of inflammatory reaction, that will lead to the initiation of cough. Here the patient may have the pricking type of pain due to the ongoing inflammatory reaction.
- 19 Itching in throat region can be seen if there is any foreign body present in the region or inflammatory reaction in the throat region. As throat is the route for respiratory system as well as alimentary canal, if the pathological event prominent in respiratory system, then the person will have cough.
- 20 This is another clinical presentation which can be associated with itching sensation in the throat region. This is specifically seen in disturbances of mouth, oral cavity, tonsils, pharynx. E.g. enlargement of tonsils. As already discussed, mouth is the route for food and air entry. That is the reason, if the person is having initiation of pathological process in this region and if it progresses towards respiratory airways, then this could be premonitory symptom for cough manifestation.
- 21 When the person is having deficiency in salivary secretion, there will be proliferation of microorganisms in the oral cavity, and these microorganisms will multiply because of the remnants of food in around the teeth. As there is reduced salivary secretion, the person will not swallow. This will lead to the coating of the tongue and palate.
- 22 The presence of hoarseness of voice is specially observed in the diseases of upper respiratory passages, larynx in particular. Any condition of nose, nasopharynx, para nasal sinuses, oropharynx will lead to the hoarseness of the voice as the normal movement of air is in the direction of the nose to larynx. Still, if the pathological events continue to progress, then the person may present with clinical features of lower respiratory tract disease. Thus, hoarseness of voice can be a premonitory symptom of respiratory disease, lower respiratory airways in particular.
- 23 As there is less salivary secretion, there is no proper contact of saliva with food during food ingestion. In other words, taste buds cannot perceive the taste because of absence of contact of saliva and food with taste buds. That is the reason, the person may have tastelessness.
- 24 This can be attributable to the activation of sympathetic nervous system as we have seen deficiency of saliva during the early respiratory disease. If the person will have continuation of pathological event, there might be a probability of stimulation of sympathetic nervous system in the stomach and distal parts also. This may lead to the reduced secretion of gastric juice, which may lead to loss of appetite.
- 25 In this type of Kasa, the cardinal feature is related to the character of the sputum. The person may have either colourless sputum or very negligible amount of sputum. The appearance of the cough is also variable. In other words, the associated features with this type of Kasa is unpredictable.

- cardiac region and head
- Hoarseness of voice
- Dryness of the chest, throat and mouth
- Horripilation
- Darkness in front of the eyes
- Hoarseness of voice on continuous coughing
- Debility
- Confusion
- Dry cough with very little amount of sputum is produced which is coming out after several bouts of stressful coughing
- Symptoms of *Vataja Kasa* going to subside after consumption of unctuous, sweet, sour, salty and warm food and drinks.

As per Sushruta and Madhavakara, the Lakshana of Vataja Kasa

- Pricking type of pain in chest, temples, head, abdomen and sides of the chest and abdomen
- Dryness of mouth
- Weak physical strength
- Hoarseness of voice
- Oja Kshaya*
- Continuous coughing as if sputum is present inside the chest, but very little sputum comes out after several bouts of coughing
- Severe hoarseness of voice after several bouts of coughing

Pittaja Kasa

कटुकोष्णविदाह्यम्लक्षाराणामति सेवनम्।
पित्तकासकरं क्रोधः संतापश्चाग्निसूर्यजः॥14॥
पीतनिष्ठीवनाक्षित्वं तिक्तास्यत्वं खरामयः।
उरोधूमायनं तृष्णा दाहो मोहोऽरुचिर्भ्रमः॥15॥
प्रततं कासमानश्च ज्योतीषीव च पश्यति।
श्लेष्माणं पित्तसंसृष्टं निष्ठीवति च पैत्तिके॥16॥ Ca.Ci. - 18/14-16
उरोविदाहज्वरवक्त्रशोषरम्यर्दितस्तिक्तमुखस्तृषार्तः।
पित्तेन पीतानि वमेत्कटूनि कासेत्सपाण्डुः परिदह्यमानः॥18॥
Su.Ut. - 52/8; Ma.Ni. - 11/6

- Excess consumption of pungent, sour taste substances
- Excess consumption of substances which are hot in potency such as chillies etc
- Excess consumption of substances which produce burning sensation in the chest and thirst
- Excess consumption of alkaline substances
- Anger

- Excess exposure to heat of fire
- Excess exposure to heat of sun

Lakshana of Pittaja Kasa²⁶

- Cough with yellowish discolouration of sputum
- Yellowish discolouration of the eyes
- Bitter taste in the mouth
- Hoarseness of voice
- Sensation of smoke coming out through chest
- Thirst
- Burning sensation
- Confusion
- Tastelessness
- Giddiness
- The person will have sensation of sparks in front of eyes after several bouts of coughing
- Sputum will be either mucoid or mucopurulent

As per Sushruta and Madhavakara,

- Burning sensation of chest
- Raised body temperature
- Dryness of mouth
- Bitter taste in the mouth
- Thirst
- Vomiting with yellowish colour in the vomitus with pungent taste in the mouth
- Patient will have *Pandu* (anemia)

Kaphaja Kasa

गुर्वभिष्यन्दिमधुरस्निग्धस्वप्नाविचेष्टनैः।
वृद्धः श्लेष्माणिलं रुद्ध्वा कफकसं करोति हि॥17॥
मन्दाग्नित्वारु चिच्छर्दि पीन सोत्क्लेशगौरवैः।
लोमहर्षास्यमाधुर्यक्लेदसं सदनैर्युतम्॥18॥
बहुलं मधुरं स्निग्धं निष्ठीवति घनं कफम्।
कासमानो ह्यरुग् वक्षः संपूर्णमिव मन्यते॥19॥ Ca.Ci. - 18/17-19
प्रलिप्यमानेन मुखेन सीदन् शिरोरुजातः कफपूर्णदेहः।
अभक्तरुगौरवसादयुक्तः कासेत ना सान्द्रकफं कफेन॥9॥
Su.Ut. - 52/9; Ma.Ni. - 11/7

- Excess consumption of substances which are heavy to digest such as blackgram items or curds etc
- Excess consumption of substances which are *Abhishyandi* in nature which are responsible for increased secretion in the channels leading to obstruction in them
- Excess consumption of sweet substances

26 In this type of Kasa, the character of the sputum plays a major role in the diagnosis as stated earlier in the context of Vataja Kasa. In case of infective respiratory conditions, the person may have either yellowish or greenish or pus mixed or reddish sputum.

- Excess consumption of unctuous substances
- Sleeping in day time in most of the times
- Physical inactivity

Lakshana of Kaphaja Kasa²⁷

- Loss of appetite
- Tastelessness
- Vomiting
- Running nose
- Nausea
- Heaviness of body parts
- Horripilation
- Sweetish taste in the mouth
- Reduced physical activity
- Nature of sputum is thick, unctuous, sweet taste in the sputum and in large amounts
- Painless coughing
- A person who hears the cough, it feels that the chest is full of sputum

As per Sushruta and Madhavakara,

- Coating over the tongue
- Headache
- Kapha will be present all over the body
- Heaviness of the body parts
- Sluggish movements of the body parts
- Thick sputum and large amount of sputum is produced

Kshataja Kasa

अतिव्यवायभाराध्ययुद्धाभ्यगजविग्रहैः।

रुक्षस्योरः क्षतं वायुर्ग्रहीत्वा कासमावहेत्॥20॥

स पूर्व कासते शुष्कं ततः धीवेत् सशोणितम्।

कण्ठेन रुजताऽत्यर्थं विरुग्णेनेव चोरसा॥21॥

सूचीभिरिव तीक्ष्णानिस्तुयमानेन शुलिना।

दुःखस्पर्शनं शुलेन भेदीडाभितापिना॥22॥

पर्वभेदज्वरश्चासतुष्णावैखर्यपीडितः।

पारावत इवाकूजन् कासवेगात्क्षतोद्भवात्॥23॥

Ca.Ci. – 18/20-23; Ma.Ni. 11/8-11

वक्षोऽतिमात्रं विहतं तु यस्य व्यायामभाराध्ययनाभिघातैः।

विश्लिष्टवक्षाः स नरः सरक्तं धीवत्यभीक्ष्णं क्षतजं तमाहुः॥10॥

Su.Ut. – 52/10

- Excess sexual intercourse
- Lifting heavy weights excessively

- Excess walk
- Fighting with physically strong people
- Falling from horse or elephant
- All above causes lead to the trauma to the chest, leads to the manifestation of cough

Lakshana of Kshataja Kasa

- Initially the patient will have dry cough which will be followed by expectoration of cough mixed with blood or haemoptysis
- Patient will have severe pain in the chest and throat
- Pricking type of pain in the chest
- Tenderness in the chest
- Severe and intolerable pain
- Joint pain, fever, breathing discomfort, thirst, hoarseness of voice
- The voice resembles the sound of pigeon while breathing

As per Sushruta,

- The chest gets injured due to excess exercise, lifting heavy weight, walking and trauma
- The person will have frank blood while coughing

Kshayaja Kasa

विषमासात्म्यभोज्यातिव्यवायाद्वेगनिग्रहात्।

घृणिनां शोचतां नृणां व्यापन्नेऽग्नौ त्रयो मलाः॥24॥

कुपिताः क्षयजं कासं कुर्युर्देहक्षयप्रदम्।

दुर्गन्धं हरितं रक्तं धीवेत् पूयोपमं कफम्॥25॥

स्थानादुत्का समानश्च हृदयं मन्यते च्युतम्।

अकस्मादुष्णाशीतार्तो बह्वाशी दुर्बलः कृशः॥26॥

स्निग्धाच्छमुखवर्णत्वम् श्रीमद्दर्शनं लोचनः।

पाणिपादतलैः श्लक्ष्णैः सततासूयको घृणी॥27॥

ज्वरो मिश्राकृतिस्तस्य पार्श्वरुक् पीनसोऽरुचिः।

भिन्नसंहतवर्चस्त्वं स्वरभेदोऽनिमित्ततः॥28॥

इत्येष क्षयजः कासः क्षीणानां देहनाशनः।

शुष्यन् विनिष्ठीवति दुर्बलस्तु प्रक्षीणमांसो रुधिरं सपूयम्॥11॥

ससर्वलिङ्गं भृशदुश्चिकित्स्यं चिकित्सितज्ञाः क्षयजं वदन्ति।

Ca.Ci. – 18/24-29

Su.Ut. – 52/11,12

विषमासात्म्यभोज्यातिव्यवायाद्वेगनिग्रहात्।

घृणिनां शोचतां नृणां व्यापन्नेऽग्नौ त्रयो मलाः।

कुपिताः क्षयजं कासं कुर्युर्देहक्षयप्रदम्॥12॥

स गात्रशूलज्वरदाहमोहान् प्राणक्षयं चोपलभेत कासी।

शुष्यन्विनिष्ठीवति दुर्बलस्तु प्रक्षीणमांसो रुधिरं सपूयम्।

तं सर्वलिङ्गं भृशदुश्चिकित्स्यं चिकित्सितज्ञाः क्षयजं वदन्ति॥13॥

Ma.Ni. – 11/12,13

G K

27

As stated in the context of Vataja and Pittaja Kasa, the characteristic features of sputum determine the Kaphaja Kasa too. In case of chronic respiratory diseases either obstructive or restrictive disease conditions, the person may have this type of sputum which may be either sticky or white or large amount of sputum.

- ✱ Consumption of food substances at improper time and quantity
- ✱ Excess consumption of unwholesome food
- ✱ Excess sexual intercourse
- ✱ Frequent suppression of natural urges
- ✱ The people who are having aversion towards any substance/ thing
- ✱ The person who is having grief continuously
- ✱ The above said causes lead to the vitiation of three *Dosha* and leading to the manifestation of *Kshayaja Kasa*.

Lakshana of Kshayaja Kasa

- ✱ The patient will have profound weight loss and deterioration of general health
- ✱ The patient will have foul smelling, greenish or reddish or pus mixed sputum
- ✱ When the person gets cough, he will have the feeling of displacement of heart from its site
- ✱ Frequent abnormal sensations of warmth and cold from the patient
- ✱ Debility
- ✱ Emaciation
- ✱ Unctuous, clear oral cavity and skin
- ✱ *Shirnat Darshana Lochana*
- ✱ Sliminess of palms and soles
- ✱ *Satata Asuyako Ghrini*
- ✱ Fever
- ✱ Pain in the sides of chest
- ✱ Running nose
- ✱ Tastelessness
- ✱ Unformed stools
- ✱ Hoarseness of voice

As per Madhavakara

- ✱ Pricking type of pain all over the body

- 28 When compared to other types of *Kasa*, *Kshayaja Kasa* is usually of longer duration and it afflicts usually physically weak individuals. In this situation, the person will have extreme *Dhatu Kshaya* in the body. That may be the reason, it may take much longer time to have complete recovery.
- 29 One of the major clinical presentation of *Kshataja Kasa* is the presence of blood in the sputum during coughing. First of all, the clinician has to ensure the exact site of bleeding either nose or nasopharynx or oropharynx or oral cavity such as teeth or larynx or lower respiratory passages. The exact site of the bleeding can be identified by proper case history and adequate physical examination. If the person is not having any major disease manifestation, then *Kshataja Kasa* can be manageable.
- 30 In *Ayurveda*, it is being said that if a person is suffering from *Kasa*, the ideal *Aushadha Sevana Kala* (time of administration of medicine is *Muhurmuhu* (frequent administration of medicine). If we make the accurate analysis of degree of *Dosha* vitiation, actual physical strength of the patient, plenty of medicines are available and the person is listening to the advices of the clinician, such type of *Kasa* is treatable.
- 31 One of the major events taking place in elderly patients having *Kasa* is *Dhatu Kshaya* (depletion of body tissue elements). Another major reason could be the elderly patients may have the association of other disease conditions also such as congestive cardiac failure or interstitial lung disease or chronic obstructive pulmonary disease. That is why, the *Kasa* appearing in elderly individuals is manageable with difficulty.
- 32 When compared to all types of *Kasa* mentioned in *Ayurveda* textbooks, these types of *Kasa* are having short duration when in disease manifestation from the time of contact of aetiological factor to the body when compared to *Kshataja* and *Kshayaja Kasa*. That may be the reason, *Doshaja* varieties of *Kasa* are easily treatable.

- ✱ Fever
- ✱ Burning sensation
- ✱ Confusion
- ✱ Life threatening
- ✱ Initially dry cough followed by haemoptysis or pus mixed sputum
- ✱ Debility
- ✱ Extreme muscle atrophy
- ✱ All symptoms observed in a single patient and it is very difficult to treat.

Sadhyasadhya of Kasa

इत्येष क्षयजः कासः क्षीणानां देहनाशनः।
साध्यो बलवतां वा स्याद्याप्यस्त्वेव क्षतोत्थितः॥14॥
नवौ कदाचित्सिद्धेतामपि पादगुणान्वितौ।
त्रीन् पूर्वासाधयेत्साध्यान्पथ्यान्प्रांस्तु यापयेत्॥15॥

Ma.Ni. – 11/14,15

- ✱ *Kshayaja Kasa*²⁸ in a physically weak individual is *Asadhya* (incurable)
- ✱ *Kshataja Kasa*²⁹ in a strong individual is *Yapya* (manageable)
- ✱ If the *Kasa* is of recent origin and if *Chikitsa Chatushpada* is adequately present, then such disease is *Krichrasadhya* (treatable with difficulty)³⁰
- ✱ In elderly people, *Kasa* is developed in old age, the *Kasa* is *Yapya* (manageable)³¹
- ✱ *Vataja*, *Pittaja* and *Kaphaja Kasa* is treatable if proper measures and *Pathya* are followed.³²

Differential diagnosis of Kasa based on Kapha (character of sputum) Lakshana Chapter resources

Character of Sputum	Vataja Kasa	Pittaja Kasa	Kaphaja Kasa
Amount	Very negligible to little	Moderate amount	Large amount

Character of Sputum	Vataja Kasa	Pittaja Kasa	Kaphaja Kasa
Colour	Colourless	Yellowish, greenish, reddish	Whitish
Duration of the illness	Very short	Chronic	Chronic

Differential diagnosis of Kshayaja Kasa and Rajayakshma

Basis	Kshayaja Kasa	Rajayakshma
Based on causes	Multiple causative factors are mentioned	In this, 4 types of Nidana exposure are mentioned: Sahasajanya, Vishamashanjanya, Sandharanajanya, Kshayajanya
Nature of Lakshana	Restricted to Pranavaha Srotas	Not restricted to Pranavaha Srotas; widespread to all Srotas
Dushya affliction	Rasa, Rakta are primarily involved	All seven Dhatu are afflicted during the pathogenesis

Basis	Kshayaja Kasa	Rajayakshma
Appearance of associated features	Bhinna Varchas (unformed stools), Svarabheda (hoarseness of voice) manifested in the individual, the cause is untraceable	The cause for the Lakshana manifested in the individual can be traceable.

Differential diagnosis of Kshataja Kasa and Kshata Kshina

Basis	Kshataja Kasa	Kshata Kshina
Duration of the illness	Acute	Chronic
Lakshana	First, there will be Shushka (dry) Kasa followed by the cough with blood	In this disease, the person will have the presence of blood along with sputum
Prognosis	Comparatively better when compared to Kshata Kshina	Comparatively poorer when compared to Kshataja Kasa

Key summary of the chapter

- ❖ कसनात्कास उच्यते Ca.Ci. – 18/8 is the Pratyatma Lakshana of Kasa observed during the manifestation of the disease.
- ❖ This disease is characterised by the production of specific type of sound i.e. sound produced by tapping the broken bronze vessel. This is facilitated by the combined action of Prana and Udana Vata.
- ❖ The disease afflicts Pranavaha Srotas. This is characterised by explosive expiration where in the irritating contents present in the Pranavaha Srotas is being eliminated through mouth. Depending upon the dominance of Dosha during the progression of the disease, we can observe the vitiation of Vata or Kapha. Udbhava Sthan of Kasa is Amashaya and Sanga and Vimargagamana is the nature of Srotodushti is seen in this disease.
- ❖ Pranavata and Udanavata are the main types of Vata Dosha responsible for the manifestation and progression of the disease. Depending upon the association of other types of Dosha, the person may have the appearance of yellowish discolouration of sputum in case of Pittaja Kasa and whitish discolouration of sputum in case of Kaphaja Kasa.

पूर्वरूपं भवेत्तेषां शुकपूर्णगलास्यता।

कण्ठे कण्डूश्च भोज्यानामवरोधश्च जायते॥5॥

Ca.Ci. – 18/5

- ❖ The sensation of pricking type with thorns in the throat and oral cavity, irritation in the throat and sensation of obstruction of food in the throat region are some of the Purvarupa (premonitory symptoms) seen in Kasa disease.
- ❖ Sankhya Samprapti of Kasa is 5 types. Vataja, Pittaja, Kaphaja, Kshataja and Kshayaja Kasa.
- ❖ As far as Sadhyasadhyata (prognosis) of the disease is concerned, Doshaja type of Kasa (Vataja, Pittaja and Kaphaja) are treatable if the person follows adequate diet and regimen. Kshataja Kasa is considered as Yapyia (manageable), the clinician should identify the exact cause of Kshataja Kasa. Kshayaja Kasa is considered as Asadhya (incurable), particularly if the person is having extreme Dhatu Kshaya.

Probable questions asked in the university examination

Long essays

- ❖ Define Kasa. Explain Nidana, Samprapti, Purvarupa, Pratyatma Lakshana, Bheda and Sadhyasadhyata of Kasa

- ④ Explain the Nidana Panchaka of Kasa

Short essays

- ④ Explain Kshataja Kasa
 ④ Describe Kshayaja Kasa
 ④ Explain Kaphaja Kasa
 ④ Explain the types of Kasa
 ④ Explain Samprapti and Samprapti Ghataka of Kasa
 ④ Describe Sadhyasadyata of Kasa

MCQs

1. Sankhya Samprapti of Kasa is

- a. 3 types; b. 4 types;
 c. 5 types; d. 7 types

Answer: (c)

2. Which of the following type of Vata involved in the Samprapti of Kasa?

- a. Pranavata; b. Samanavata;
 c. Apanavata; d. Vyanavata

Answer: (a)

3. Which of the following type/s of Vata involved in the Samprapti of Kasa?

- a. Pranavata;
 b. Udanavata;
 c. Both a and b;
 d. None of the above

Answer: (c)

4. 'Paravata Iva Kujan' is the term specifically seen in which type of Kasa?

- a. Kshayaja Kasa;
 b. Kshataja Kasa;
 c. Pittaja Kasa;
 d. Vataja Kasa

Answer: (b)

5. Which of the following type of Kasa is considered as Asadya in nature?

- a. Vataja Kasa;
 b. Pittaja Kasa;
 c. Kaphaja Kasa;
 d. Kshayaja Kasa

Answer: (d)

6. Shukapurna Galasyata is of Kasa.

- a. Nidana; b. Purvarupa;
 c. Rupa; d. Upadrava

Answer: (b)

7. Jyotishamlva Darshana is the Lakshana observed in

- a. Vataja Kasa; b. Pittaja Kasa;
 c. Kaphaja Kasa; d. Kshayaja Kasa

Answer: (b)

8. Uro Dhumayana is the Lakshana observed in

- a. Pittaja Kasa;
 b. Purvarupa of Chardi;
 c. Amlapitta;
 d. Annaja Hikka

Answer: (a)

9. Kantha Kandu is the Purvarupa of which of the following?

- a. Kasa; b. Shvasa;
 c. Hikka; d. Galaganda

Answer: (a)

10. Kshataja Kasa is

- a. Sukhasadhyaya; b. Krichrasadhyaya;
 c. Yasya; d. Pratyakhyeya

Answer: (c)

Chapter resources

- ④ <https://pharmaceutical-journal.com/article/ld/case-based-learning-cough>
 ④ Charaka Samhita, Chikitsa Sthana 18th chapter – Kasa Chikitsa Adhyaya
 ④ Sushruta Samhita Uttara Tantra 52nd chapter – Kasa Pratishedha Adhyaya
 ④ Sushruta Samhita Uttara Tantra 50th chapter – Hikka Pratishedha Adhyaya
 ④ Madhava Nidana 11th chapter – Kasa Nidana Adhyaya
 ④ Golwalla's medicine for students by Aspi F Golwalla and Sharukh A Golwalla, edited by Milind Y Nadkar, 25th edition, Jaypee Brothers Medical Publishers (P) Ltd, New Delhi, 2017
 ④ The Merck Manual of Diagnosis and Therapy by Robert S Porter and Justin L Kaplan, 19th edition, Merck Sharp & Dohme Corp, New Jersey, 2011
 ④ Chamberlain's symptoms and signs in clinical medicine by Colin Ogilvie and Christopher C Evans, 12th edition, Butterworth-Heinemann Publishers, 1997
 ④ Best & Taylor's physiological basis of medical practice by O P Tandon and Y Tripathi, 13th edition, Wolters Kluwer Health (India) Pvt. Ltd, Gurgaon, 2012
 ④ Textbook of Medical physiology by Arthur C Guyton and John E Hall, 11th edition, Saunders Elsevier publications, Philadelphia, 2006